LUAD vs. LUSC Subtyping Using TransMIL Framework

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COMP411/511 PROJECT PRESENTATION

Problem Definition

Challenge: Subtyping Lung Adenocarcinoma (LUAD) and Lung Squamous Cell Carcinoma (LUSC) from Whole Slide Images (WSIs).

Why it Matters: Accurate subtyping impacts treatment decisions and patient outcomes.

Key Issues:

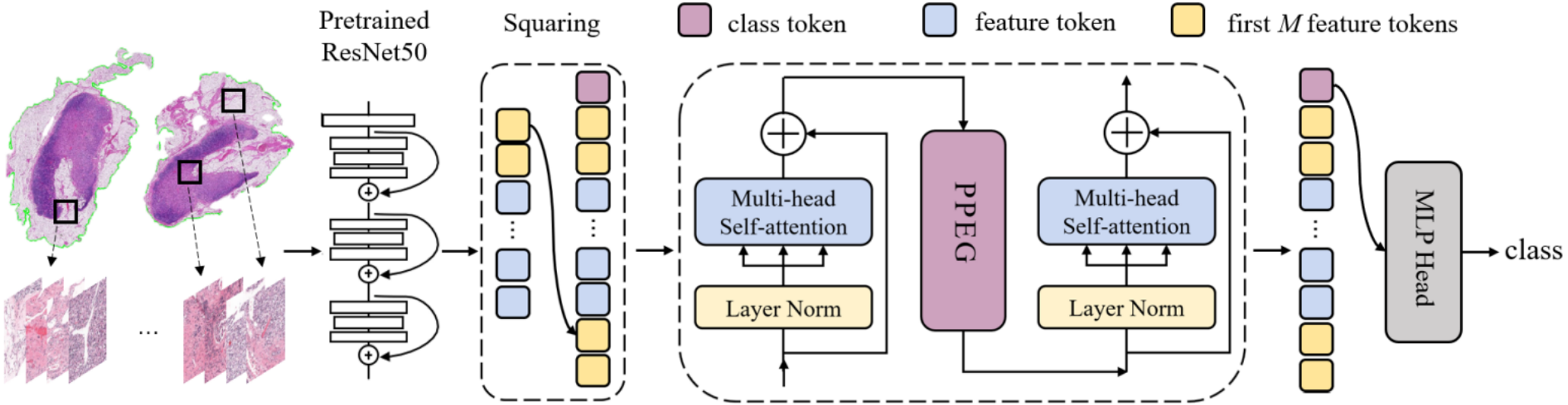
- WSIs are enormous, containing billions of pixels.
- Slide-level labels only; no region-level annotations.
- Contextual and spatial relationships are critical for diagnosis.
- Requires a **Multiple Instance Learning (MIL) framework** to handle slide-level labels and aggregate patch-level information effectively.

cross-entropy loss
optimizer : Lookahead
learning rate : 2×10−4
weight decay : 1×10−5
mini-batch size : 1

Baseline and Methods

Baseline Model: TransMIL with ResNet50 as feature extractor.

Improvements: Feature extraction with UNI Encoder



[6] Zhuchen Shao, Hao Bian, Yang Chen, Yifeng Wang, Jian Zhang, Xiangyang Ji, and Yongbing Zhang. Transmil: Transformer based correlated multiple instance learning for whole slide image classification. In Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS). NeurIPS, 2021.
[8]Michael Ruogu Zhang, James Lucas, Geoffrey E. Hinton, and Jimmy Ba. Lookahead optimizer: k steps forward, 1 step back. In Advances in Neural Information Processing Systems, pages 9597–9608, 2019.

Training & Evaluation

Dataset:

- TCGA-NSCLC: 201 WSIs (109 LUAD, 92 LUSC).
- Slide-level labels only; open-access samples used.

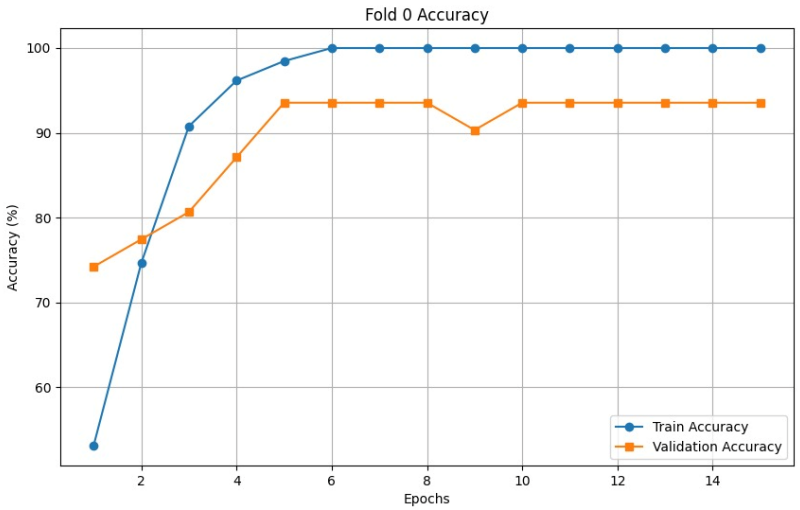
Split: 65-20-15 train-validation-test split.

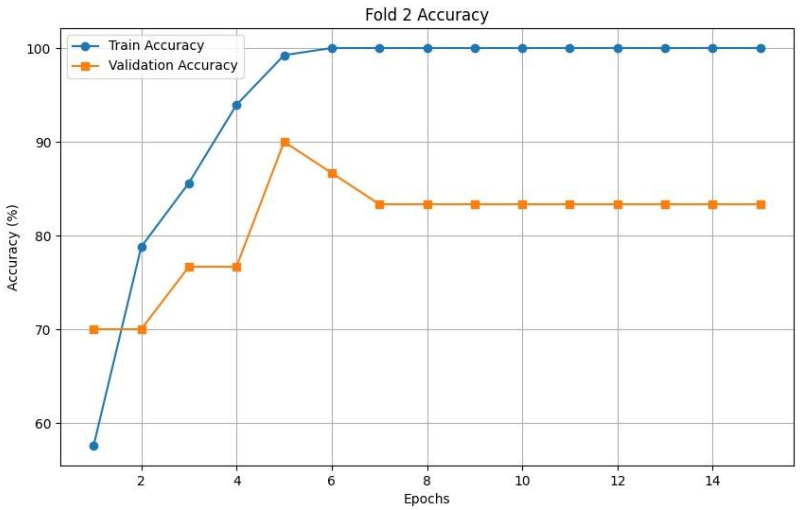
Validation: 5-fold cross-validation for robustness.

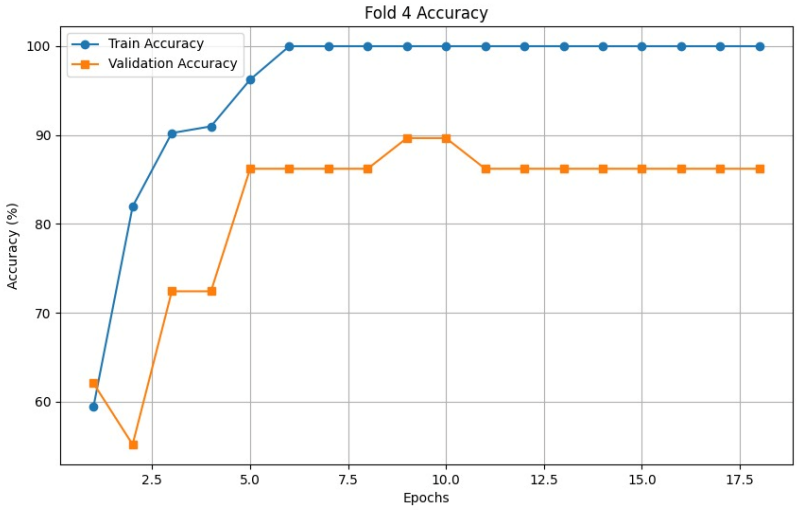
Metrics: Accuracy, Precision, Recall, F1-score.

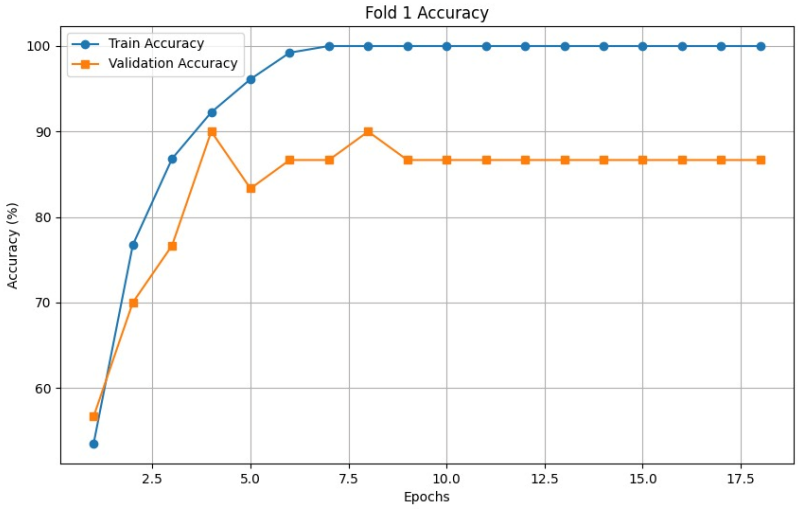
The Cancer Genome Atlas Research Network. Comprehensive molecular profiling of lung adenocarcinoma. *Nature*. 2014; 511(7511):543–550. doi:10.1038/nature13385.

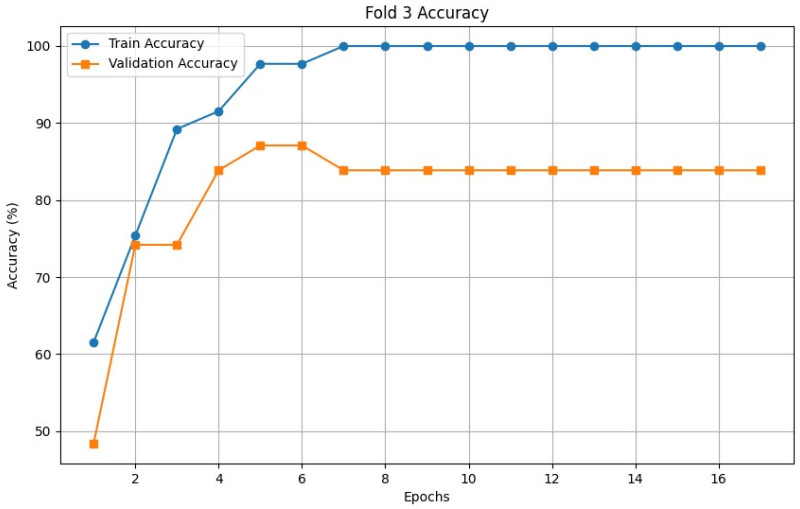
Training and Validation Accuracy Plots Across Epochs











Results Overview

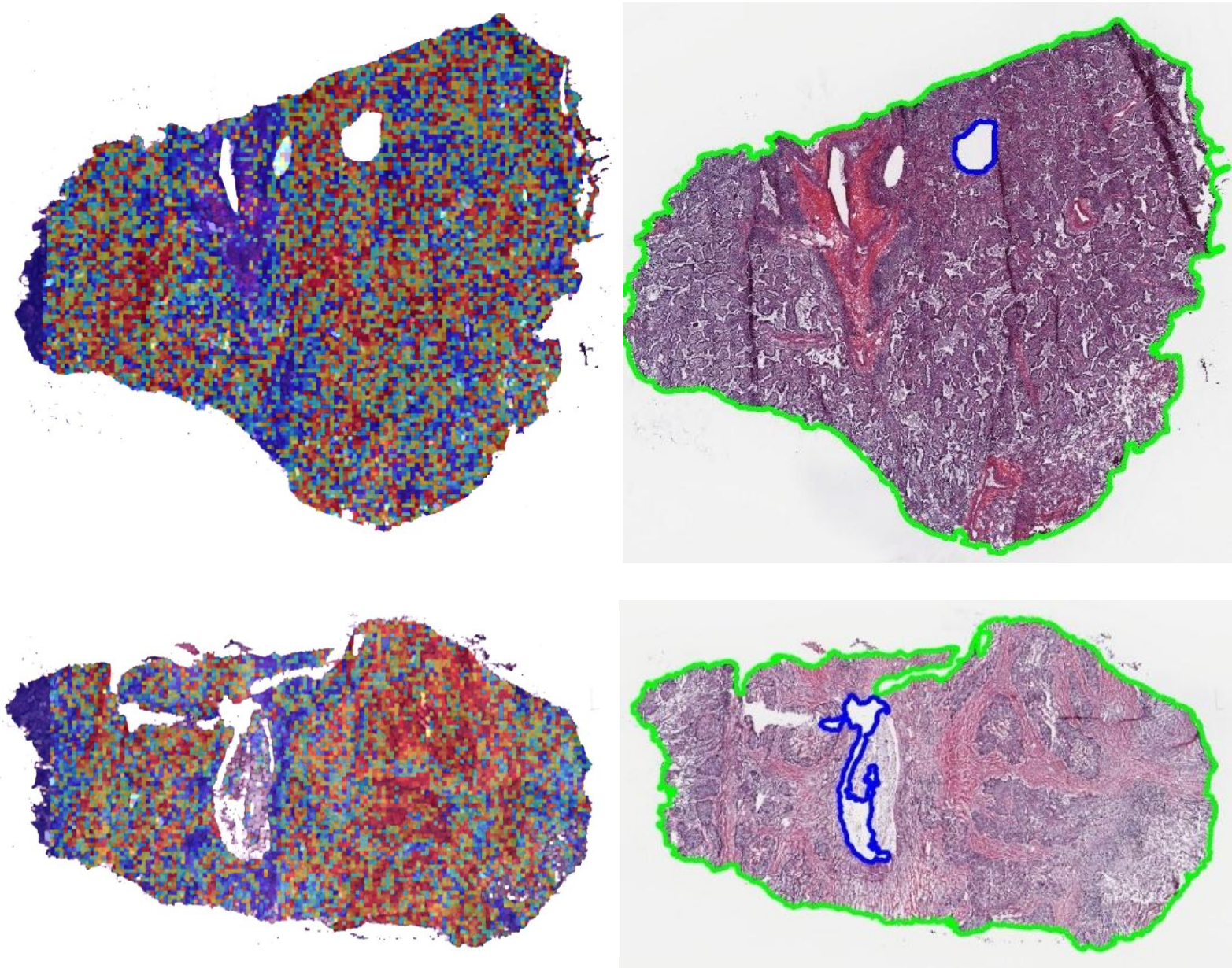
UNI Encoder Results

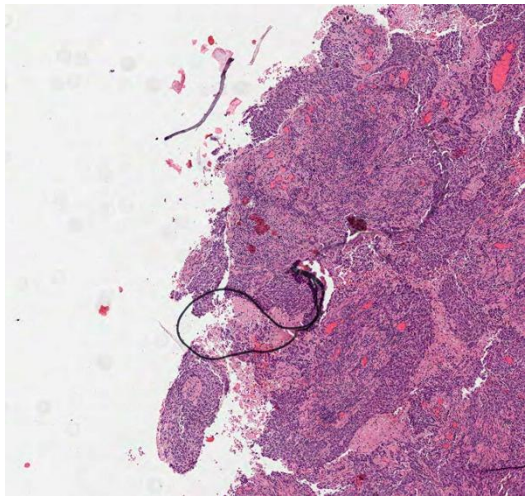
Accuracy | F1 Score	87.38 | 87.64
Precision | Recall	91.00 | 85.06
LUSC Accuracy | LUAD Accuracy	90.12 | 85.06

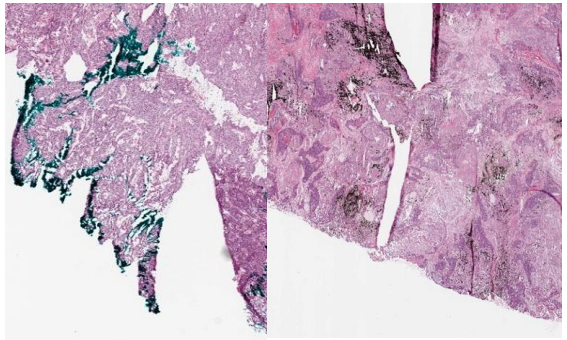
ResNet50 Results

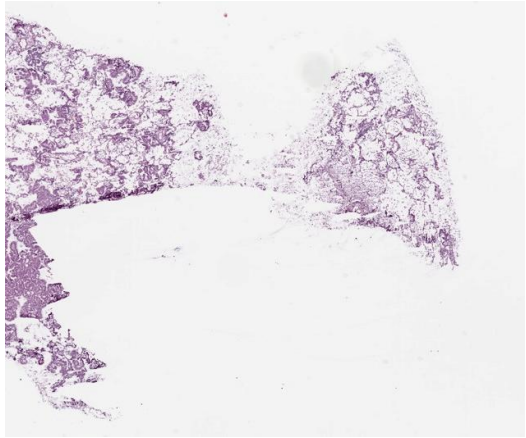
Accuracy | F1 Score	72.98 | 72.87
Precision | Recall	79.07 | 71.67
LUSC Accuracy | LUAD Accuracy	74.74 | 71.67

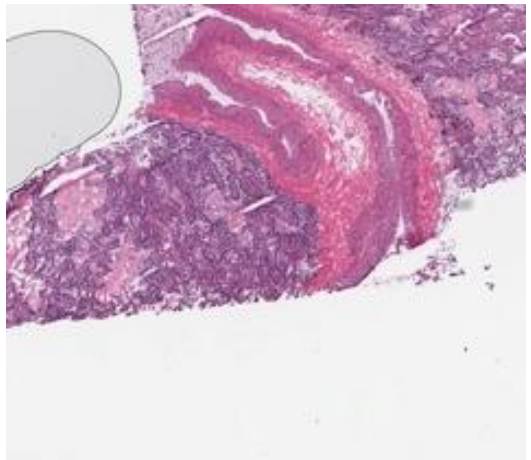
Visual Interpretability











Challenges and Lessons Learned

Challenges:

- Time-consuming dataset download and preprocessing.
- Modifications to adapt heatmap generation for transformers.
- Storage and dependency issues.

Lessons Learned:

- Importance of domain-specific feature extractors.

Conclusion and Future Work

Key Takeaways:

- UNI Encoder significantly outperformed ResNet50 for LUAD/LUSC classification.
- Domain-specific pretraining enhances medical image analysis.

Future Directions:

- Expand dataset with additional WSIs.
- Validate heatmaps clinically with pathologists.
- Explore other transformer-based architectures.

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